CHAPTER 2 Biochemistry

SYLLABUS

Biomolecules— Structure and Functions; Biological Membranes— Structure, Action Potential and Transport Processes; Enzymes — Classification, Kinetics and Mechanism of Action; Basic Concepts and Designs of Metabolism (Carbohydrates, Lipids, Amino Acids and Nucleic Acids) Photosynthesis, Respiration and Electron Transport Chain; Bioenergetics.

CHAPTER ANALYSIS

Topic	GATE 2010	GATE 2011	GATE 2012	GATE 2013	GATE 2014	GATE 2015	GATE 2016	GATE 2017	GATE 2018	GATE 2019
Biomolecules: Structure and Functions	2	1	2	2	3	2	2	5	2	3
Biological Membranes: Structure, Action Potential and Transport Processes	1		2	1	2		1			1
Enzymes: Classification, Kinetics and Mechanism of Action	3	1	4	5		3	2	2		2
Basic Concepts and Designs of Metabolism: (Carbohydrates, Lipids, Amino Acids and Nucleic Acids) Photosynthesis, Respiration and Electron Transport Chain	5	2	4		3		1		2	
Bioenergetics	1			1	1	2			1	1

QUESTIONS

1. During lactic acid fermentation, net yield of ATP and NADH per mole of glucose is

(A) 2 ATP and 2 NADH.
(B) 2 ATP and 0 NADH.
(C) 4 ATP and 2 NADH.
(D) 4 ATP and 0 NADH.

(GATE 2010; 1 Mark)

2. Identify the enzyme that catalyzes the following reaction.

$$\alpha\text{-Ketoglutarate} + NADH + NH_4^+ + H^+ \rightarrow \text{Glutamate} + NAD^+ + H_2O$$

(A) Glutamate synthetase
(B) Glutamate oxoglutarate aminotransferase
(C) Glutamate dehydrogenase
(D) α-Ketoglutarate deaminase

(GATE 2010; 1 Mark)

3. The degree of inhibition for an enzyme catalyzed reaction at a particular inhibitor concentration is independent of initial substrate concentration. The inhibition follows

(A) competitive inhibition.
(B) mixed inhibition.
(C) uncompetitive inhibition.
(D) non-competitive inhibition.

(GATE 2010; 1 Mark)

4. Oxidation reduction reactions with positive standard redox potential (ΔE°) have

(A) positive ΔG°.
(B) negative ΔG°.
(C) positive ΔE^+.
(D) negative ΔE^+.

(GATE 2010; 1 Mark)

5. Match the chemicals in Group I with the possible type/class in Group II.

Group I	Group II
P. Pichwazu	1. Vitamin
Q. Zeatin	2. Auxin
R. Thiamine	3. Amino Acid
S. Glutamine	4. Cytokinin

(A) P–2; Q–4; R–1; S–3
(B) P–3; Q–1; R–2; S–3
(C) P–3; Q–1; R–2; S–4
(D) P–4; Q–2; R–1; S–3

(GATE 2010; 2 Marks)

6. The turnover numbers for the enzymes E1 and E2 are 150 s^{-1} and 15 s^{-1} respectively. This means

(A) E1 binds to its substrate with higher affinity than E2.
(B) The velocity of reactions catalyzed by E1 and E2 at their respective saturating substrate concentrations could be equal, if concentration of E2 used is 10 times that of E1.
(C) The velocity of E1 catalyzed reaction is always greater than that of E2.
(D) The velocity of E1 catalyzed reaction at a particular enzyme concentration and saturating substrate concentration is lower than that of E2 catalyzed reaction under the same conditions.

(GATE 2010; 2 Marks)

7. The dissociation constant K_d for ligand binding to the receptor is 10^{-7} M. The concentration of ligand required for occupying 10% of receptors is

(A) 10^{-8} M
(B) 10^{-7} M
(C) 10^{-6} M
(D) 10^{-5} M

(GATE 2010; 2 Marks)

Common Data for Questions 8 and 9: The width of the lipid bilayer membrane is 30 Å. It is permeated by a protein which is a right handed α-helix.

8. The number of α-helical turns permeating the membrane is

(A) 5.6 turns
(B) 3.5 turns
(C) 6.5 turns
(D) 5.0 turns

(GATE 2010; 2 Marks)

9. The number of amino acid residues present in the protein is

(A) 15
(B) 18
(C) 17
(D) 20

(GATE 2010; 2 Marks)

Statement for Linked Answer Questions 10 and 11: The standard redox potential values for two half-reactions are given below. The value for Faraday's constant is 96.48 kJ V^{-1} mol^{-1} and Gas constant R is 8.31 J K^{-1}, mol^{-1}.

$NAD^+ + H^+ + 2e^- \leftrightarrow NADH$ $\quad -0.315$ V

$FAD + 2H^+ + 2e^- \leftrightarrow FADH_2$ $\quad -0.219$ V

10. The ΔG° for the oxidation of NADH by FAD is

(A) -9.25 kJ mol^{-1}
(B) -103.04 kJ mol^{-1}
(C) $+51.52$ kJ mol^{-1}
(D) -18.5 kJ mol^{-1}

(GATE 2010; 2 Marks)

11. The value of ΔG°, if given K_{eq} is 1.7, at 23°C will be

(A) -17.19 kJ mol^{-1}
(B) -19.8 kJ mol^{-1}
(C) $+52.82$ kJ mol^{-1}
(D) -117.07 kJ mol^{-1}

(GATE 2010; 2 Marks)

12. An alternative to glycolysis pathway is

(A) glyoxylate pathway.
(B) pentose phosphate pathway.
(C) citric acid cycle.
(D) gluconeogenesis.

(GATE 2011; 1 Mark)

13. In *N*-linked glycosylation, the oligosaccharide chain is attached to protein by

(A) asparagine.
(B) arginine.
(C) serine.
(D) threonine.

(GATE 2011; 1 Mark)

14. Match items in Group I with Group II.

Group I	Group II
P. Glycolytic pathway	1. Chloroplast
Q. Eukaryotic oxidative metabolism	2. Glyoxysomes
R. Glyoxylate cycle	3. Mitochondria
S. Calvin cycle	4. Cytosol

(A) P–1; Q–2; R–3; S–4
(B) P–2; Q–3; R–4; S–1
(C) P–4; Q–3; R–2; S–1
(D) P–3; Q–4; R–1; S–2

(GATE 2011; 2 Marks)

15. Determine the correctness or otherwise of the following Assertion (*A*) and the Reason (*R*).

Assertion: A very low amount of inhibitor can act as an activator for allosteric enzymes.

Reason: Allosteric enzymes follow Michaelis–Menten kinetics.

(A) Both (*A*) and (*R*) are true and (*R*) is the correct reason for (*A*).
(B) Both (*A*) and (*R*) are true but (*R*) is not the correct reason for (*A*).
(C) (*A*) is true but (*R*) is false.
(D) (*A*) is false but (*R*) is true.

(GATE 2011; 2 Marks)

16. During photorespiration under low CO_2 and high O_2 levels, O_2 reacts with ribulose 1,5- bisphosphate to yield
(A) one molecule each of 3-phosphoglycerate and 2-phosphoglycolate.
(B) two molecules of 3-phosphoglycerate.
(C) two molecules of 2-phosphoglycolate.
(D) one molecule each of 3-phosphoglycerate and glyoxylate.

(GATE 2012; 1 Mark)

17. The activity of an enzyme is expressed in International Units (IU). However, the SI unit for enzyme activity is Katal. One Katal is
(A) 1.66×10^4 IU (B) 60 IU
(C) 6×10^7 IU (D) 10^6 IU

(GATE 2012; 1 Mark)

18. Identify the statement that is not applicable to an enzyme catalyzed reaction.
(A) Enzyme catalysis involves propinquity effects.
(B) The binding of substrate to the active site causes a strain in the substrate.
(C) Enzymes do not accelerate the rate of reverse reaction.
(D) Enzyme catalysis involves acid-base chemistry.

(GATE 2012; 1 Mark)

19. The molarity of water in a water: ethanol mixture (15 : 85, v/v) is approximately
(A) 0.85 (B) 5.55
(C) 8.5 (D) 55.5

(GATE 2012; 1 Mark)

20. In zinc finger proteins, the amino acid residues that coordinate zinc are
(A) Cys and His (B) Asp and Glu
(C) Arg and Lys (D) Asp and Arg

(GATE 2012; 2 Marks)

21. Determine the correctness or otherwise of the following Assertion (*A*) and Reason (*R*).
Assertion: Plants convert fatty acids into glucose.
Reason: Plants have peroxisomes.
(A) Both (*A*) and (*R*) are true but (*R*) is not the correct reason for (*A*).
(B) Both (*A*) and (*R*) are true and (*R*) is the correct reason for (*A*).
(C) (*A*) is true but (*R*) is false.
(D) (*A*) is false but (*R*) is true.

(GATE 2012; 2 Marks)

22. Match the high energy compounds in Group I with the biosynthetic pathways for the molecules in Group II.

Group I	Group II
P. GTP	1. Fatty acid
Q. UTP	2. Phospholipid
R. CTP	3. Protein
S. Acyl coenzyme A	4. Peptidoglycan

(A) P–3; Q–2; R–4; S–1
(B) P–2; Q–4; R–3; S–1
(C) P–4; Q–3; R–1; S–2
(D) P–3; Q–4; R–2; S–1

(GATE 2012; 2 Marks)

23. Match the vitamins in Group I with the processes/ reactions in Group II.

Group I	Group II
P. Pantothenic acid	1. Electron transport
Q. Vitamin B_2	2. Transfer of 1-C units
R. Vitamin B_6	3. Decarboxylation
S. Folic acid	4. Fatty acid metabolism
	5. Hydrolysis

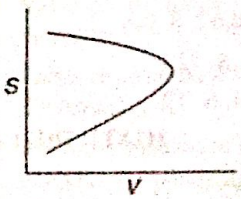
(A) P–5; Q–2; R–4; S–1
(B) P–4; Q–1; R–3; S–2
(C) P–4; Q–2; R–1; S–3
(D) P–2; Q–1; R–3; S–5

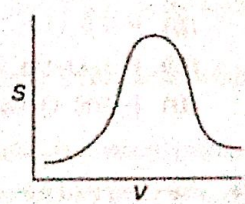
(GATE 2012; 2 Marks)

24. An enzymatic reaction is described by the following rate expression.

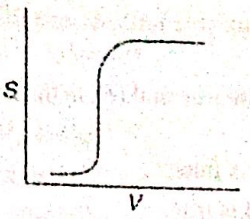
$$v = \frac{V_m S}{K_M + S + S^2 / K_s}$$

Which one of the following curves represents this expression?

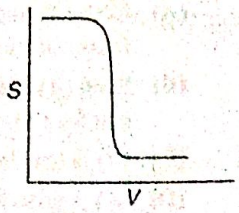
(A)


(B)


(C)



(D)



(GATE 2012; 2 Marks)

Common Data for Questions 25 and 26: In a muscle, the extracellular and intracellular concentrations of Na^+ are 150 mM and 12 mM, and those of K^+ are 2.7 mM and 140 mM, respectively. Assume that the temperature is 25° C and that the membrane potential is –60 mV, with the interior more negatively charged than the exterior. (R = 8.314 J mol^{-1} K^{-1}; F = 96.45 kJ mol^{-1} V^{-1})

25. The free energy change for the transport of three Na^+ out of the cell is

(A) +1.5 kJ mol^{-1} (B) +17.4 kJ mol^{-1}

(C) +18.9 kJ mol^{-1} (D) +36.3 kJ mol^{-1}

(GATE 2012; 2 Marks)

26. The free energy change for the transport of two K^+ into the cell is

(A) +8.0 kJ mol^{-1} (B) +11.6 kJ mol^{-1}

(C) +19.6 kJ mol^{-1} (D) +31.2 kJ mol^{-1}

(GATE 2012; 2 Marks)

Common Data for Questions 27 and 28: The purification data for an enzyme is given below:

	Step	Volume (mL)	Total protein (mg)	Total activity (Units)	Specific activity (Units/mg)
P	Cell-free extract	17	177	102	0.58
Q	Q-Sepharose	14	18.8	72	3.83
R	Phenyl Sepharose	26	9.2	45	4.89
S	Sephacryl S-200	7	4.1	30	7.32

27. The fold purification for each step is

(A) P–0.1, Q– 0.66, R–0.84, S–1.26

(B) P–1.0, Q–0.52, R–0.67, S–0.8

(C) P–1, Q–6.6, R–8.4, S–12.6

(D) P–100, Q–66, R–84, S–12

(GATE 2012; 2 Marks)

28. The yield (%) for each step is

(A) P–10, Q–7.2, R–4.5, S–2.0

(B) P–34, Q–24, R–15, S–1

(C) P–3.4, Q–2.4, R–1.5, S–0.1

(D) P–100, Q–71, R–44, S–29

(GATE 2012; 2 Marks)

29. Which one of the following is an ABC transporter?

(A) Multidrug resistance protein

(B) Acetylcholine receptor

(C) Bacteriorhodopsin

(D) ATP synthase

(GATE 2013; 1 Mark)

30. The catalytic efficiency for an enzyme is defined as

(A) k_{cat} (B) $\frac{V_{max}}{k_{cat}}$

(C) $\frac{k_{cat}}{K_M}$ (D) $\frac{k_{cat}}{V_{max}}$

(GATE 2013; 1 Mark)

31. Match the entries in Group I with the enzymes in Group II.

Group I	Group II
P. NAD^+	1. Glutathione peroxidase
Q. Selenium	2. Nitrogenase
R. Pyridoxal phosphate	3. Lactate dehydrogenase
S. Molybdenum	4. Glycogen phosphorylase

(A) P–3; Q–2; R–4; S–1

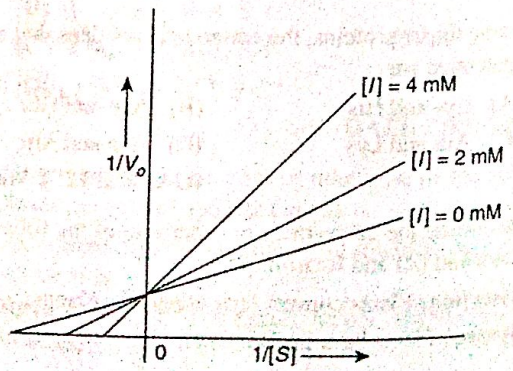
(B) P–4; Q–1; R–3; S–2

(C) P–3; Q–1; R–4; S–2

(D) P–3; Q–4; R–2; S–1

(GATE 2013; 2 Marks)

32. The activity of an enzyme was measured by varying the concentration of the substrate (S) in the presence of three different concentrations of inhibitor (I) 0, 2 and 4 mM. The double reciprocal plot given below suggests that the inhibitor (I) exhibits



(A) substrate inhibition.

(B) uncompetitive inhibition.

(C) mixed inhibition.
(D) competitive inhibition.

(GATE 2013; 2 Marks)

Common Data for Questions 33 and 34: A solution was prepared by dissolving 100 mg of protein X in 100 mL of water. Molecular weight of protein X is 15,000 Da; Avogadro's number = 6.022×10^{23}.

33. Calculate the molarity (μM) of the resulting solution.

(A) 66.6 (B) 6.6
(C) 0.67 (D) 0.067

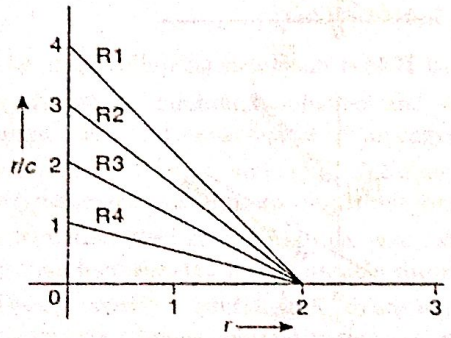
(GATE 2013; 2 Marks)

34. The number of molecules present in this solution is

(A) 40.15×10^{19} (B) 6.023×10^{19}
(C) 4.015×10^{19} (D) 0.08×10^{19}

(GATE 2013; 2 Marks)

Common Data for Questions 35 and 36: The binding efficiency of three different receptors R1, R2 and R3 were tested against a ligand using equilibrium dialysis, with a constant concentration of receptor and varying concentrations of ligand. The Scatchard plot of receptor titration with different concentration of ligand is given below (r is moles of bound ligand per moles of receptor and c is concentration of free ligand).



35. The number of ligand binding sites present on receptors R1 and R3, respectively are

(A) 1 and 4 (B) 1 and 1
(C) 4 and 1 (D) 2 and 2

(GATE 2013; 2 Marks)

36. Which one of the receptors has the highest affinity for the ligand?

(A) R1 (B) R2
(C) R3 (D) R4

(GATE 2013; 2 Marks)

37. The product(s) resulting from the hydrolysis of maltose is/are

(A) a mixture of α-D-Glucose and β-D-Glucose.
(B) a mixture of D-Glucose and L-Glucose.
(C) α-D-Glucose only.
(D) β-D-Glucose only.

(GATE 2014; 1 Mark)

38. Amino acid residue which is most likely to be found in the interior of water-soluble globular proteins is

(A) threonine (B) aspartic acid
(C) valine (D) histidine

(GATE 2014; 1 Mark)

39. The reactions leading to the formation of amino acids from the TCA cycle intermediates are

(A) carboxylation. (B) isomerization
(C) transamination. (D) decarboxylation.

(GATE 2014; 1 Mark)

40. Triose phosphate isomerase converts dihydroxy acetone phosphate (DHAP) to glyceraldehyde-3-phosphate (G-3-P) in a reversible reaction. At 298 K and pH 7.0, the equilibrium mixture contains 40 mM DHAP and 4 mM G-3-P. Assume that the reaction started with 44 mM DHAP and no G-3-P. The standard free-energy change in kJ mol^{-1} for the formation of G-3-P [$R = 8.315$ J mol^{-1} K^{-1}] is _____.

(GATE 2014; 2 Marks)

41. Match the following photoreceptors with their prosthetic groups and spectral specificity

Photoreceptor	Moiety that absorbs light	Absorption (nm)
P. Phototropin	1. Chromobilin	a. 400–500
Q. Cryptochrome	2. FAD	b. 600–800
R. Phytochrome	3. FMN	c. 500–600

(A) P–3 –a; Q–2–a; R–1–b
(B) P–1–b; Q–1 –a; R–3–b
(C) P–3–a; Q–1–a; R–1–c
(D) P–2–c; Q–1–c; R–1–a

(GATE 2014; 2 Marks)

42. If a plant is shifted to cold temperature, which of the following changes would take place in its membrane?

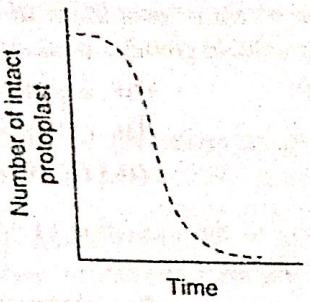
(A) Ratio of unsaturated to saturated fatty acids would increase.
(B) Ratio of unsaturated to saturated fatty acids would decrease.
(C) Absolute amount of both fatty acids would increase keeping the ratio same.
(D) Absolute amount of both fatty acids would remain unchanged.

(GATE 2014; 2 Marks)

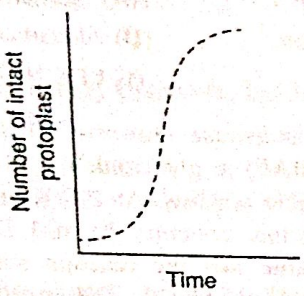
43. If protoplasts are placed in distilled water, they swell and burst as a result of endosmosis.

The plot representing the kinetics of burst is

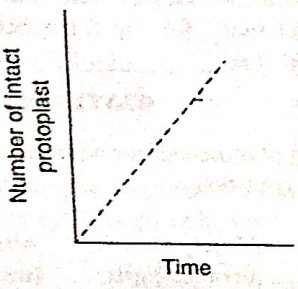
(A)



(B)



(C)



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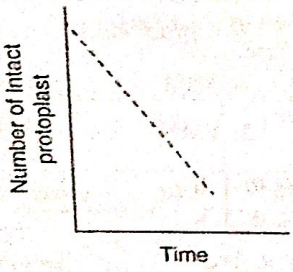


(GATE 2014; 2 Marks)

44. The active site in the alpha/beta barrel structures is usually located

(A) inside the barrel.
(B) at the amino side of the strands.
(C) at the carboxy side of the strands.
(D) at any arbitrary site.

(GATE 2014; 2 Marks)

45. Levinthal's paradox is related to

(A) protein secretion. **(B)** protein degradation.
(C) protein folding. **(D)** protein trafficking.

(GATE 2015; 1 Mark)

46. A single subunit enzyme converts 420 μ moles of substrate to product in one minute. The activity of the enzyme is _____ $\times 10^{-6}$ Katal.

(GATE 2015; 1 Mark)

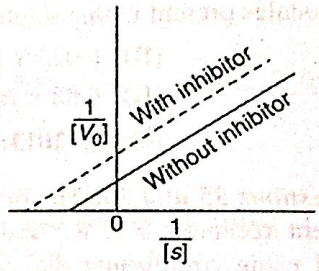
47. Which one of the following amino acids has the highest probability to be found on the surface of a typical globular protein in aqueous environment?

(A) Ala **(B)** Val
(C) Arg **(D)** Ile

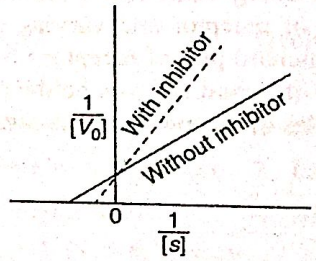
(GATE 2015; 1 Mark)

48. Which one of the following graphs represents uncompetitive inhibition?

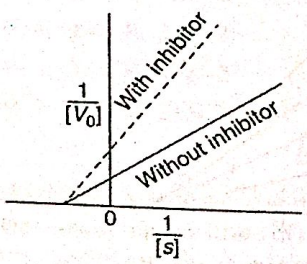
(A)



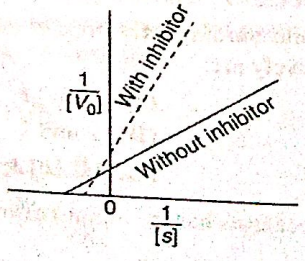
(B)



(C)



(D)



(GATE 2015; 2 Marks)

49. The K_i of a novel competitive inhibitor designed against an enzyme is 2.5 μM. The enzyme was assayed in the absence or presence of the inhibitor (5 μM) under identical conditions. The K_M in the presence of the inhibitor was found to be 30 μM. The K_M in the absence of the inhibitor is _____ μM.

(GATE 2015; 2 Marks)

50. The standard free energy change ($\Delta G'^{\circ}$) for ATP hydrolysis is –30 kJ·mole^{-1}. The *in vivo* concentrations of ATP, ADP and P_i in *E. coli* are 7.90, 1.04 and 7.90 mM, respectively. When *E. coli* cells are cultured at 37°C, the free energy change (ΔG) for ATP hydrolysis *in vivo* is _____ kJ mol^{-1}.

(GATE 2015; 2 Marks)

51. ATP biosynthesis takes place utilizing the H^+ gradient in mitochondria and chloroplasts. Identify the correct sites of H^+ gradient formation.

(A) Across the outer membrane of mitochondria and across the inner membrane of chloroplast.

(B) Across the inner membrane of mitochondria and across the thylakoid membrane of chloroplast.

(C) Within the matrix of mitochondria and across the inner membrane of chloroplast.

(D) Within the matrix of mitochondria and within the stroma of chloroplast.

(GATE 2016; 1 Mark)

52. Disaccharide molecules that contain $\beta(1 \rightarrow 4)$ glycosidic linkage are

(A) sucrose and maltose.

(B) sucrose and isomaltose.

(C) maltose and isomaltose.

(D) lactose and cellobiose.

(GATE 2016; 1 Mark)

53. Which one of the following statements is NOT true?

(A) In competitive inhibition, substrate and inhibitor compete for the same active site of an enzyme.

(B) Addition of a large amount of substrate to an enzyme cannot overcome uncompetitive inhibition.

(C) A transition state analogue in enzyme catalyzed reaction increases the rate of product formation.

(D) In non-competitive inhibition, K_M of an enzyme for its substrate remains constant as the concentration of the inhibitor increases.

(GATE 2016; 1 Mark)

54. The equilibrium potential of a biological membrane for Na^+ is 55 mV at 37°C. Concentration of Na^+ inside the cell is 20mM. Assuming the membrane is permeable to Na^+ only, the Na^+ concentration outside the membrane will be _____ mM.

(Faraday constant: 23062 cal V^{-1} mol^{-1}, Gas constant: 1.98 cal mol^{-1} K^{-1})

(GATE 2016; 2 Marks)

55. In an assay of the type II dehydroquinase of molecular mass 16 kDa, it is found that the V_{max} of the enzyme is 0.0134 μmol min^{-1} when 1.8 μg enzyme is added to the assay mixture. If the K_M for the substrate is 25 μM, the k_{cat}/K_M ratio will be _____ $\times 10^6$ M^{-1} s^{-1}.

(GATE 2016; 2 Marks)

56. Analysis of a hexapeptide using enzymatic cleavage reveals the following result:

- Amino acid composition of the peptide is: 2R, A, V, S, Y
- Trypsin digestion yields two fragments and the compositions are: (R, A, V) and (R, S, Y)
- Chymotrypsin digestion yields two fragments and the compositions are: (A, R, V, Y) and (R, S)
- Digestion with carboxypeptidase A yields no cleavage product.

Given: Trypsin cleaves at carboxyl side of R.
Chymotrypsin cleaves at carboxyl side of Y.
Carboxypeptidase A cleaves at amino side of the C-terminal amino acid (except R and K) of the peptide.
The correct amino acid sequence of the peptide is:

(A) RSYRVA **(B)** AVRYSR
(C) SRYVAR **(D)** SVRRYA

(GATE 2016; 2 Marks)

57. An enzyme catalyzes a reaction by

(A) decreasing the energy of the substrate.

(B) decreasing the activation energy of the reaction.

(C) decreasing product stability.

(D) increasing the activation barrier of the reaction.

(GATE 2017; 1 Mark)

58. Natural proteins are composed primarily of 20 α-amino acids. Which one of the following statements is true for any of these amino acids in a solution of pH 1.5?

(A) Only the amino group is ionized.

(B) Only the carboxylic acid group is ionized.

(C) Both amino and carboxylic acid groups are ionized.

(D) Both amino and carboxylic add groups are neutral.

(GATE 2017; 1 Mark)

59. If a protein contains four cysteine residues, the number of different ways they can simultaneously form two intramolecular disulfide bonds is _____.

(GATE 2017; 1 Mark)

60. An enzymatic reaction exhibits Michaelis–Menten kinetics. For this reaction, on doubling the concentration of enzyme while maintaining $[S] >> [E_t]$,

(A) both K_M and V_{max} will remain the same.

(B) K_M will remain the same but V_{max} will increase.

(C) K_M will increase but V_{max} will remain the same.

(D) both K_M and V_{max} will increase.

(GATE 2017; 1 Mark)

61. Which one of the following amino acids has three ionizable groups?

(A) Glycine (B) Leucine
(C) Valine (D) Lysine

(GATE 2017; 2 Marks)

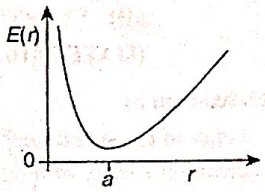
62. If the chemical composition of proteins in an organism is $CH_{1.5}O_{0.3}N_{0.3}S_{0.004}$ the mass percentage of carbon in the proteins is _____.

Given data: Atomic weights (Da) of C = 12, H = 1, O = 16, N = 14, and S = 32.

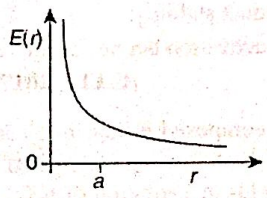
(GATE 2017; 2 Marks)

63. The interaction energy E between two spherical particles is plotted as a function of the distance (r) between them. When $r < a$, where a is a constant, the net force between the spherical particles is repulsive. When $r \geq a$, they attract via van der Waals attraction. Which one of the following plots correctly represents the interaction energy between the above two particles?

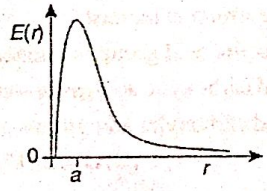
(A)



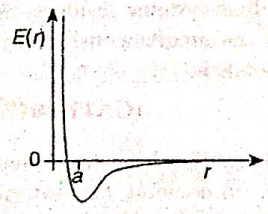
(B)



(C)



(D)



(GATE 2017; 2 Marks)

64. Which of the following components constitute a molecular mechanics force field?

P. Bond stretching Q. Bond angle bending
R. Torsional bond rotation S. Non-bonded interactions

(A) P and Q only (B) P, Q and R only
(C) P, Q and S only (D) P, Q, R and S

(GATE 2018; 1 Mark)

65. Which one of the following is INCORRECT about protein structures?

(A) A protein fold is stabilized by favorable non-covalent interactions.
(B) All parts of a fold can be classified as helices, strands or turns.
(C) Two non-covalent atoms cannot be closer than the sum of their van der Waals radii.
(D) The peptide bond is nearly planar.

(GATE 2018; 1 Mark)

66. Which one of the following metabolic processes in mammalian cells does NOT occur in the mitochondria?

(A) Citric acid cycle
(B) Oxidative phosphorylation
(C) Fatty acid β-oxidation
(D) Glycolysis

(GATE 2018; 1 Mark)

67. Yeast converts glucose to ethanol and carbon dioxide by glycolysis as per the following reaction:

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$$

Assuming complete conversion, the amount of ethanol produced (in g) from 200 g of glucose is (up to two decimal places)_____.

(GATE 2018; 2 Marks)

68. Consider a simple uni-substrate enzyme that follows Michaelis-Menten kinetics. When the enzyme catalyzed reaction was carried out in the presence of 10 nM concentration of an inhibitor, there was no change in the maximal velocity. However, the slope of the Lineweaver-Burk plot increased 3-fold. The dissociation constant for the enzyme-inhibitor complex (in nM) is _____.

(GATE 2018; 2 Marks)

69. A rod-shaped bacterium has a length of 2 µm, diameter of 1 µm and density the same as that of water. If proteins constitute 15% of the cell mass and the average protein has a mass of 50 kDa, the number of proteins in the cell is_____.

(1 Da = 1.6×10^{-24}g)

(GATE 2018; 2 Marks)

70. The mass of 1 kmol of oxygen molecules is _____ g (rounded off to the nearest integer).

(GATE 2019; 1 Mark)

71. Which one of the following need NOT be conserved in a biochemical reaction?

(A) Total mass
(B) Total moles
(C) Number of atoms of each element
(D) Total energy

(GATE 2019; 1 Mark)

72. Which one of the following statements is CORRECT for enzyme catalyzed reactions? (ΔG is Gibbs free energy change, K_{eq} is equilibrium constant)

(A) Enzymes affect ΔG, but not K_{eq}
(B) Enzymes affect K_{eq}, but not ΔG
(C) Enzymes affect ΔG and K_{eq}
(D) Enzymes do not affect ΔG or K_{eq}

(GATE 2019; 1 Mark)

73. Protein concentration of a crude enzyme preparation was 10 mg mL^{-1}. 10 µL of this sample gave an activity of 5 µmol min^{-1} under standard assay conditions. The specific activity of this crude enzyme preparation is _____ units mg^{-1}.

(GATE 2019; 1 Mark)

74. Which one of the following amino acid residues will destabilize an α-helix when inserted in the middle of the helix?

(A) Pro
(B) Val
(C) Ile
(D) Leu

(GATE 2019; 2 Marks)

75. In general, which one of the following statements is NOT CORRECT?

(A) Hydrogen bonds result from electrostatic interactions.
(B) Hydrogen bonds contribute to the folding energy of proteins.
(C) Hydrogen bonds are weaker than van der Waals interactions.
(D) Hydrogen bonds are directional.

(GATE 2019; 2 Marks)

76. The difference in concentrations of an uncharged solute between two compartments is 1.6-fold. The energy required for active transport of the solute across the membrane separating the two compartments is _____ cal mol^{-1} (rounded off to the nearest integer). (R = 1.987 cal mol^{-1} K^{-1}, T = 37°C).

(GATE 2019; 2 Marks)

ANSWER KEY

1. (B)	2. (C)	3. (D)	4. (B)	5. (A)	6. (C)
7. (C)	8. (A)	9. (D)	10. (D)	11. (A)	12. (B)
13. (A)	14. (C)	15. (C)	16. (A)	17. (C)	18. (C)
19. (C)	20. (A)	21. (A)	22. (D)	23. (B)	24. (A)
25. (D)	26. (D)	27. (C)	28. (D)	29. (A)	30. (C)
31. (C)	32. (D)	33. (66.6 µM)	34. (4.01×10^{10})	35. (D)	36. (A)
37. (A)	38. (C)	39. (C)	40. (5.7 kJ mol^{-1})	41. (A)	42. (A)
43. (A)	44. (C)	45. (C)	46. (7×10^{-6} Katal)	47. (C)	48. (A)
49. (10 µM)	50. (−47.7 kJ mol^{-1})	51. (B)	52. (D)	53. (C)	54. (158.49 mM)
55. (8.93×10^{5} M^{-1} s^{-1})	56. (B)	57. (B)	58. (A)	59. (3)	60. (B)
61. (D)	62. (53.03%)	63. (D)	64. (D)	65. (B)	66. (D)
67. (102.22 g)	68. (5 nM)	69. (2.94×10^{6})	70. (32000 g)	71. (B)	72. (A)
73. (50 µ mol min^{-1} mg^{-1})	74. (A)	75. (C)	76. (289.51 cal mol^{-1})		

ANSWERS WITH EXPLANATION

1. Topic: Metabolism—Fermentation

Lactic acid fermentation is observed in some bacteria. It is also observed in the muscles of animals, wherein, during vigorous activity, when the demand for ATP is high and oxygen is in short supply, ATP is mainly synthesized via anaerobic glycolysis. In this process, ATP is generated rapidly as compared to the slower process of oxidative respiration.

Under these conditions, the enzyme lactate dehydrogenase (LDH) catalyzes the oxidation of NADH by pyruvate to yield NAD^+ and lactic acid.

$$\text{Pyruvic acid} + \text{NADH} + H^+ \xrightarrow{\text{Lactic dehydrogenase, } Mg^{2+}, \text{FMN}} \text{Lactic acid} + NAD^+$$

Under anaerobic conditions only 2 molecules of ATP are produced (as a result of glycolysis).

Answer (B)

2. Topic: Metabolism—Amino Acids

Glutamate dehydrogenase (GDH) is located in the mitochondria. GDH catalyzes the reversible $NADP^+$-linked oxidative deamination of L-glutamate into alpha ketoglutarate and ammonia in two steps.

- The first step involves a Schiff base intermediate being formed between ammonia and alpha ketoglutarate. This Schiff base intermediate is crucial because it establishes the alpha carbon atom in glutamate's stereochemistry.
- The second step involves the Schiff base intermediate being protonated, which is done by the transfer of a hydride ion from NADPH resulting in L-glutamate.

Answer (C)

3. Topic: Enzymes—Kinetics

In non-competitive inhibition, inhibitors do not compete with the substrate for the same binding site of the enzyme; rather, the inhibitor binds to a site other than the enzyme's active site. This changes the conformation of the active site in such a manner that the substrate can no longer bind to the active site. This type of inhibition depends only on the inhibitor concentration and is independent of the concentration of the substrate. This is because in the presence of a non-competitive inhibitor, a certain fraction of the enzyme molecules is necessarily inactive at any given instant. The inactivation of cytochrome oxidase by cyanide is an example of non-competitive inhibition.

Answer (D)

4. Topic: Bioenergetics

A redox reaction is spontaneous, if the standard electrode potential for the redox reaction is positive. And, for a spontaneous redox reaction, Gibbs free energy, ΔG° must be negative, in accordance with the following equation:

$$\Delta G^\circ = -nFE^\circ$$

Answer (B)

5. Topic: Biomolecules—Structure and Function

- Picloram is a synthetic auxin or also called as auxin mimic. It is used as herbicide to kill susceptible plants by imitating the plant growth hormone auxin (indole acetic acid). On its administration in effective doses, it can cause uncontrolled and disorganized growth of plant that leads to its death.
- Zeatin is the natural cytokinin found in corn kernels and coconut milk. Zeatin was first isolated from young corn seed endosperm in 1963. It is the most common cytokinin in plants.
- Thiamine is Vitamin B_1. It acts as coenzyme for many different enzymes that break carbon-to-carbon bonds and are involved in carbohydrate metabolism of pyruvic acid to CO_2 and H_2O. It is essential for synthesis of neurotransmitter acetylcholine. Its deficiency in body causes Beri-Beri, which is partial paralysis of smooth muscle of alimentary canal, causing digestive disturbances and skeletal muscle paralysis.
- Glutamine is a standard neutral amino acid. It is found abundantly in muscles and gets converted into glutamic acid essential for cerebral function. It helps to treat arthritis, autoimmune diseases, fibrosis, etc.

Answer (A)

6. Topic: Enzymes—Kinetics

The turnover number of an enzyme is the number of reaction processes (turnovers) that each active site catalyzes per unit time. It is also defined as the catalytic constant, k_{cat} of an enzyme as:

$$k_{cat} = \frac{V_{max}}{[E]_T}$$

Here, k_{cat} = Turnover number,

V_{max} = Maximum rate of reaction when all the enzyme catalytic sites are saturated with substrate and

E_T = Total enzyme concentration or concentration of total enzyme catalytic sites.

Thus, with this relation, we know that for the same concentration of both the enzyme, velocity of E1 will be greater than that of E2, as turnover number of E1 is higher.

Answer (C)

7. Topic: Enzymes—Kinetics

$$\underset{(L)}{\text{Ligand}} + \underset{(R)}{\text{Receptor}} \underset{K_d}{\overset{K_a}{\rightleftharpoons}} [LR]$$

$$\underset{\text{Association constant}}{K_a} = \frac{[LR]}{[L][R]} \text{ or } \underset{\text{Dissociation constant}}{K_d} = \frac{[L][R]}{[LR]}$$

Given, 10 % receptor is occupied,

$$K_d = 10^{-7}\ M$$

$$[L] + 0.9[R] \rightleftharpoons 0.1[LR]$$

$$[L] = \frac{K_d \times [LR]}{[R]}$$

$$= \frac{0.1 \times 10^{-7}}{0.9} \approx 10^{-8}$$

Answer (C)

8. Topic: Biological Membrane—Structure

We know,

In a right handed α-helix, in a pitch of 5.1Å, number of turn = 1

Given, Width of the lipid bilayer membrane = 30 Å

For 30 Å, number of turns = 30/5.1 = 5.6 turns

Answer (A)

9. Topic: Biomolecules—Structure of Proteins

We know,

In a right handed α-helix, number of residues in 1 turn = 3.6

Number of residues in 5.6 turns = 5.6×3.6

$= 20$

Answer (D)

10. Topic: Electron Transport Chain

Given, F = is 96.48 kJ V^{-1} mol^{-1}

$R = 8.31\ J\ K^{-1}\ mol^{-1}$

$NAD^+ + H^+ + 2e^- \longleftrightarrow NADH \quad -0.315\ V$

$FAD^+ + 2H^+ + 2e^- \longleftrightarrow FADH_2 \quad -0.219\ V$

Oxidation of NADH by FAD

$$\Delta E^\circ_{cell} = E_{red} - E_{ox}$$

$$= -0.219 - (-0.315)$$

$$= 0.096\ V$$

$$\Delta G^\circ = -nF\Delta E^\circ_{cell}$$

$$= -2 \times 96.48 \times 0.096$$

$$= -18.5\ kJ\ mol^{-1}$$

Answer (D)

11. Topic: Electron Transport Chain

Given, $K_{eq} = 1.7$

Temperature, $T = 23°C = 23 + 273 = 296\ K$

$\Delta G = ?$

$$\Delta G = \Delta G^\circ + RT \ln K_{eq}$$

$$= -18.5 + 8.31 \times 296 \times \ln 1.7$$

$$= -17.2\ kJ\ mol^{-1}$$

Answer (A)

12. Topic: Metabolism—Glycolysis

The pentose phosphate pathway or phosphogluconate pathway or hexose monophosphate shunt is a metabolic pathway alternate to glycolysis. It produces NADPH and pentoses (5-carbon sugars) as well as ribose 5-phosphate, the last one is a precursor for the synthesis of nucleotides.

Answer (B)

13. Topic: Biomolecules—Structure

N-linked glycosylation or *N* glycosylation refers to the attachment of glycan (oligosaccharide) to the amide (N-group) of asparagine of the protein.

Answer (A)

14. Topic: Metabolism—Carbohydrates

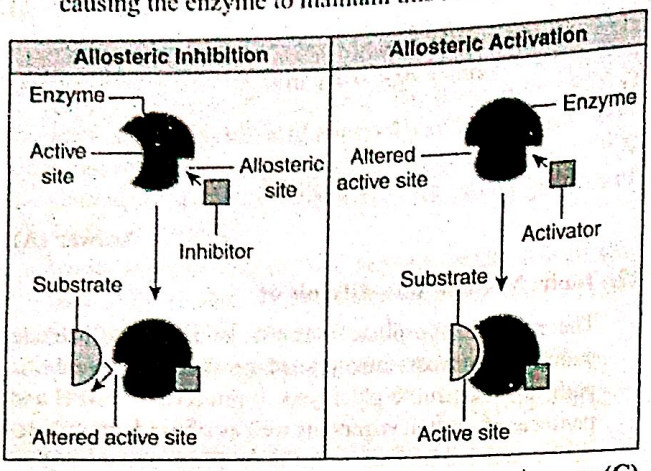
- Glycolysis or glycolytic pathway converts glucose to pyruvate through a series of intermediate metabolites which occurs in cytoplasm.
- Eukaryotic oxidative metabolism occurs at the internal folded mitochondrial membranes called cristae.
- The glyoxylate cycle occurs in plant's glyoxysomes (special peroxisomes). This cycle lets seeds to use lipids as a source of energy to form the shoot system during germination.
- The Calvin cycle (C_3 cycle) is the set of chemical reactions that occurs in chloroplasts during photosynthesis. The reaction is light-independent because it takes place after the energy has been trapped from sunlight.

Answer (C)

15. Topic: Enzymes—Kinetics

An allosteric enzyme is an enzyme that contains a region to which small, regulatory molecules (called effectors) may bind in addition to and separate from the substrate binding site and thereby affect the catalytic activity. Allosteric enzymes are an exception to the Michaelis–Menten model. Because they have more than two subunits and active sites, they do not obey the Michaelis–Menten kinetics, but instead have sigmoidal kinetics. Allosteric

enzymes have two states: a low affinity state dubbed the T state and the high affinity R state. Inhibitors work by preferentially binding to the T state of an allosteric enzyme, causing the enzyme to maintain this low affinity state.



Answer (C)

16. Topic: Photorespiration

Photorespiration is process in which RuBisCO oxygenates RuBP rather than addition of carbon dioxide. It decreases the efficiency of photosynthesis and reduced the photosynthetic output by 25%. During photorespiration, oxygen is added to ribulose-1, 5-bisphosphate to produce one molecule of 3-phosphoglycerate and one molecule of 2-phosphoglycolate.

Answer (A)

17. Topic: Enzymes—Mechanism of Action

Enzyme activity of any enzyme is expressed in terms of international unit. One international unit of enzyme activity can be defined as amount of enzyme consuming or producing 1 micro-mole of substrate or product per minute under standard conditions. In contrast, Katal is the unit and which gives the amount of enzyme required to convert 1 mol of substrate per second.

$$\text{Enzyme unit (IU)} = \frac{\mu\text{mol}}{\text{min}} = \frac{10^{-6}\,\text{mol}}{60\,\text{sec}} = \frac{1}{6\times10^{7}}\,\text{Katal}$$

$$\therefore\ 1\,\text{Katal} = 6\times10^{7}\,\text{IU}$$

Answer (C)

18. Topic: Enzymes—Mechanism of Action

Enzymes are biological catalyst which speed up the rate of reaction by lowering the activation energy of the system. They show proximity and propinquity effects and being protein in nature they often show acid-base chemistry. They do not change the equilibrium constant and free energy of the system. Thus, they catalyze both forward as well as backward reactions in order to achieve e juilibrium state.

Answer (C)

19. Topic: Biomolecules

Molarity is defined as number of moles of solute per liter of solution

Molarity = (weight of solute/mol. mass of solute) / (1000/vol. of solution in mL)

Weight of water = 15mL = 15g (taking density of water to be 1)

Molecular mass of water (H_2O) = 2 + 16 = 18

Volume of solution = 100 mL

Hence, molarity = (15/18) × (1000/100)

= 0.833 × 10 = 8.33 M

Answer (C)

20. Topic: Biomolecules—Protein Structure and Function

Zinc finger proteins are among the most abundant proteins in eukaryotic genomes. Their functions are extraordinarily diverse and include DNA recognition, RNA packaging, transcriptional activation, regulation of apoptosis, protein folding and assembly and lipid binding. The zinc ion is coordinated by two histidine residues and two cysteine residues.

Answer (A)

21. Topic: Metabolism—Lipids

Fatty acid degradation in most organisms occurs through the beta-oxidation cycle. In mammals, beta-oxidation occurs in both mitochondria and peroxisomes, whereas in plants and most fungi the beta-oxidation cycle occurs only in the peroxisomes. In plant cells, peroxisomes play a variety of roles including converting fatty acids to sugar and assisting chloroplasts in photorespiration. In animal cells, peroxisomes protect the cell from its own production of toxic hydrogen peroxide.

Answer (A)

22. Topic: Metabolism

- GTP-binding protein regulators are proteins that regulate the function of small G proteins.
- In the step of peptidoglycan synthesis, the N-acetyl-glucosamine-1-phosphate attacks UTP (Uridine triphosphate), a pyrimidine nucleotide, has the ability to act as an energy source.
- Two minor phospholipids are phosphatidylglycerol (PG) and phosphatidylinositol (PI). They are derived from phosphatidic acid instead of 1,2-diglyceride. Phosphatidic acid is activated by activation using CTP to form the high energy compound CDP-diglyceride.
- Acyl coenzyme A is a group of coenzymes involved in the metabolism of fatty acids.

Answer (D)

23. Topic: Metabolism

- Pantothenate (vitamin B_5) is the key precursor for the biosynthesis of coenzyme A (CoA), an essential

cofactor involved in a many metabolic reactions, including the synthesis of phospholipids, the synthesis and degradation of fatty acids, and the operation of the tricarboxylic acid cycle (TCA or Krebs' cycle).

- FAD and FMN are called as flavins, since they are derived from riboflavin (Vitamin B_2). Flavoproteins of electron transport chain (ETC), including FMN in Complex I and FAD in Complex II.
- Vitamin B_6 (pyridoxine) is a water-soluble vitamin that is naturally present in many foods. It helps in carboxylation reactions.
- Tetrahydrafolate (THF) is derived from the vitamin folic acid (folate). We eat folate and use the enzyme dihydrofolate reductase to convert it into tetrahydrofolate, which is the active form that carries 1-carbon groups in a variety of reactions.

Answer (B)

24. Topic: Enzymes—Kinetics

Given,

$$v = \frac{V_m S}{K_M + S + S^2 / K_s}$$

It can also be written as:

$$v = \frac{V_m S}{K_M + S\left(1 + S / K_s\right)}$$

This is factor $\left(1 + \frac{S}{K_s}\right)$ of the substrate inhibition, S_i. In substrate inhibition, the velocity of enzymatic reaction increases with increase in substrate concentration, till velocity reaches v_m. After achieving maximum velocity, any further increase in substrate concentration, starts decreasing the velocity. Thus, only graph of option (A) is satisfying the condition.

Answer (A)

25. Topic: Biological Membrane—Action Potential

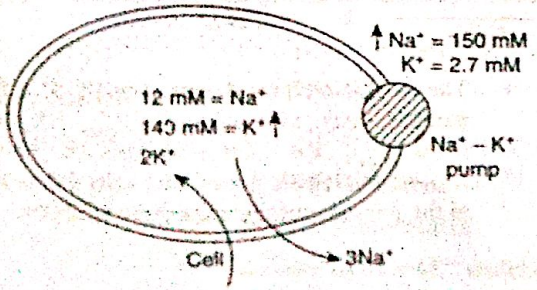
Given,

Temperature, $T = 25°C = 25 + 273 = 298$ K

Membrane potential, $\Delta\psi = -60$ mV $= -60 \times 10^{-3}$ V (As interior is more negatively charged than exterior, we can take this value as $+60 \times 10^{-3}$ V for calculation)

$R = 8.314$ J mol^{-1} K^{-1}

$F = 96.45$ kJ mol^{-1} V^{-1} $= 96.45 \times 10^3$ J mol^{-1} V^{-1}



Free energy change for transport of one Na^+ ions from inside to outside that is against the concentration gradient, must be positive. It can be calculated as:

$$\Delta G_{Na^+} = RT \ln \frac{[Na^+]_{out}}{[Na^+]_{in}} + Z \cdot F \cdot \Delta\psi$$

$$= 8.314 \times 298 \times \ln \frac{150\,\text{mM}}{12\,\text{mM}} + 1 \times 96.45 \times 10^3 \times (60 \times 10^{-3})$$

$$= 2477.572 \times \ln 12.5 - 5787$$

$$= 2477.572 \times 2.526 - 5787$$

$$= 6257.674 + 5787$$

$$= 12044.674 \text{ J mol}^{-1} = 12.04 \text{ kJ mol}^{-1}$$

For 3 Na^+ ions, it will be three times, thus

$$\Delta G_{3Na^+} = 12.04 \times 3$$

$$= 36.12 \text{ kJ mol}^{-1}$$

Answer (D)

26. Topic: Biological Membrane—Action Potential

Free energy change for transport of one K^+ ions from outside to inside that is against the concentration gradient, must be positive. It can be calculated as:

$$\Delta G_{K^+} = RT \ln \frac{[K^+]_{in}}{[K^+]_{out}} + Z \cdot F \cdot \Delta\psi$$

$$= 8.314 \times 298 \times \ln \frac{140\,\text{mM}}{2.7\,\text{mM}} + 1 \times 96.45 \times 10^3 \times (60 \times 10^{-3})$$

$$= 2477.575 \times \ln 51.85 + 5787$$

$$= 2477.575 \times 3.95 + 5787$$

$$= 9779.94 + 5787$$

$$= 15566.94 \text{ J mol}^{-1} = 15.57 \text{ kJ mol}^{-1}$$

For 2 K^+ ions, it will be two times, thus

$$\Delta G_{2K^+} = 15.57 \times 2$$

$$= 31.14 \text{ kJ mol}^{-1}$$

Answer (D)

27. Topic: Enzymes—Kinetics

Fold purification can be calculated at each step by dividing specific activity of each step by initial specific activity. Therefore,

$$\text{At P, Fold purification} = \frac{0.58}{0.58} = 1$$

$$\text{At Q, Fold purification} = \frac{3.83}{0.58} = 6.6$$

$$\text{At R, Fold purification} = \frac{4.89}{0.58} = 8.43$$

At S, Fold purification $= \frac{7.32}{0.58} = 12.62$

Answer (C)

28. Topic: Enzymes—Kinetics

For percentage yield calculation at each step, divide total enzyme activity at each step by starting total activity multiply by 100.

At P, Percentage yield $= \frac{102}{102} \times 100 = 100\%$

At Q, Percentage yield $= \frac{72}{102} \times 100 = 70.59\%$

At R, Percentage yield $= \frac{45}{102} \times 100 = 44.12\%$

At S, Percentage yield $= \frac{30}{102} \times 100 = 29.41\%$

Answer (D)

29. Topic: Biological Membrane—Transport Process

ABC (ATP-binding cassette) transporters are the members of membrane transport system which use ATP to transport material across the plasma membrane. They play crucial role in developing multidrug resistance in a cell. They do so by actively pumping out the drug of the cell.

Answer (A)

30. Topic: Enzyme—Kinetics

The catalytic efficiency or specificity constant refers to the measure of efficiency of an enzyme, i.e., how efficiently an enzyme can convert a substrate into the product.

It can be calculated as k_{cat}/K_M.

where, k_{cat} is a turnover number of an enzyme, i.e. no. of times each enzyme site converts a substrate into its product in unit time, and K_M is a Michaelis constant which represents the substrate concentration at which rate of reaction is half of maximum velocity (1/2 V_{max}).

Answer (C)

31. Topic: Bioenergetics

- Lactate dehydrogenase (LDH) is an enzyme required during the process of turning sugar into energy for cells. LDH or LD is found in nearly all living cells including animals, plants, and prokaryotes. It catalyzes the conversion of lactate to pyruvic acid and reverse, as it converts NAD^+ to NADH and reverse too.
- Glutathione peroxidase or GPx is a selenium-containing antioxidant enzyme that catalyzes the reduction of hydrogen peroxide (H_2O_2) to water and oxygen and in turn oxidizes glutathione to glutathione disulfide.
- Glycogen phosphorylase is an enzyme that catalyzes the phosphorolysis of glycogen to produce glucose 1-phosphate, having a cofactor called pyridoxal-5'-phosphate, that is located in the active site and bound to a K681 residue with a Schiff base linkage. It shuttles the phosphate group onto the substrate.
- Nitrogenases are enzymes that are produced by certain bacteria. These enzymes are responsible for the reduction of nitrogen (N_2) to ammonia (NH_3). It is composed of mainly two component proteins called the Fe protein and the Mo-Fe protein.

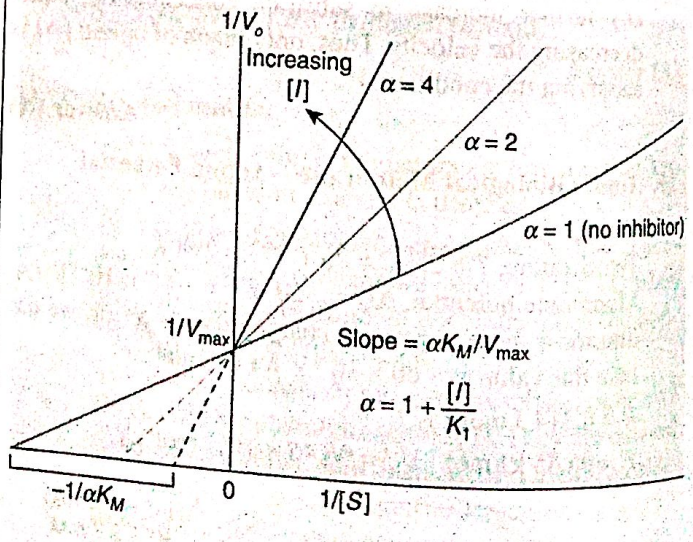
Answer (C)

32. Topic: Enzymes—Kinetics

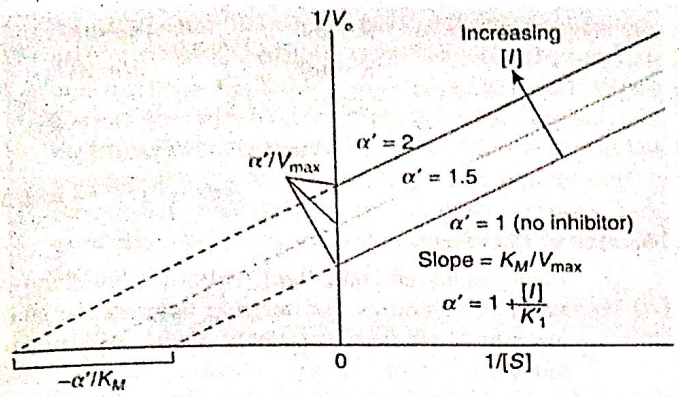
A substance that competes directly with a normal substrate for an enzyme's substrate-binding site is known as a competitive inhibitor. Such an inhibitor usually resembles the substrate so that it specifically binds to the active site but differs from the substrate so that it cannot react as the substrate does. Its equation will be:

$$\frac{1}{V_o} = \left(\frac{\alpha K_M}{V_{max}}\right)\frac{1}{[S]} + \frac{1}{V_{max}}$$

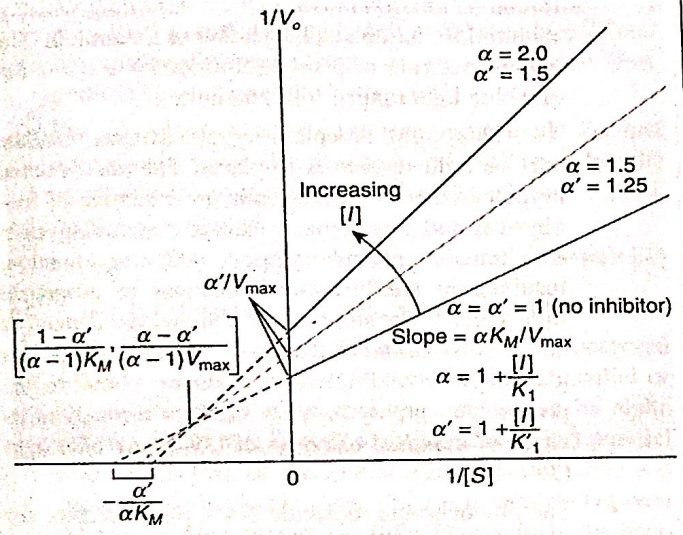
A plot of this equation is linear and has a slope of $\alpha K_M/V_{max}$, a $1/[S]$ intercept of $-1/\alpha K_M$, and a $1/V_o$ intercept of $1/V_{max}$. The double-reciprocal plots for a competitive inhibitor at various concentrations of I intersect at $1/V_{max}$ on the $1/V_o$ axis, a property that is diagnostic of competitive inhibition.



- The double-reciprocal plot consists of a family of parallel lines with slope K_M/V_{max}, $1/V_o$ intercepts of α'/V_{max}, and $1/[S]$ intercepts of $-\alpha'/K_M$ is a Lineweaver–Burk plot of Michaelis–Menten enzyme in the presence of uncompetitive inhibitor.



- In a Lineweaver–Burk plot of Michaelis–Menten enzyme in the presence of mixed inhibitor, lines intersect to the left of the $1/V_o$ axis



Answer (D)

33. Topic: Biomolecules

Given, Weight of protein = 100 mg = 0.1 g

Molecular weight of protein = 15000 Da = 15×10^3 g

Volume = 100 mL = 0.1 L

$N_A = 6.022 \times 10^{23}$

$$\text{Molarity}(\mu M) = \frac{\text{Weight of protein}}{\text{Molecular weight of protein}}$$

$$= \frac{0.1}{15 \times 10^3} = 66.6 \mu M$$

Answer (66.6 μM)

34. Topic: Biomolecules

Number of molecules = $N_A \times$ Molarity

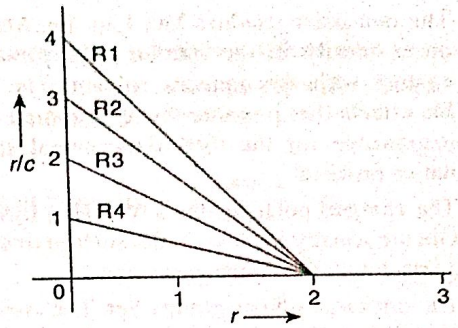
$= 6.022 \times 10^{23} \times 66.6 \times 10^{-6}$

$= 4.01 \times 10^{19}$ molecules

Answer (4.01×10^{19} molecules)

35. Topic: Enzymes—Kinetics

A receptor has a limited number of binding sites and is therefore saturated at high ligand concentration. As



shown in the plot between ratio of moles of bound ligand per moles to the concentration of free ligand (r/c) and moles of bound ligand per moles (r), is same starting from the same point for all the receptors. Thus, all the receptors (R1, R2 and R3) must have same number of two binding sites for ligand.

Answer (D)

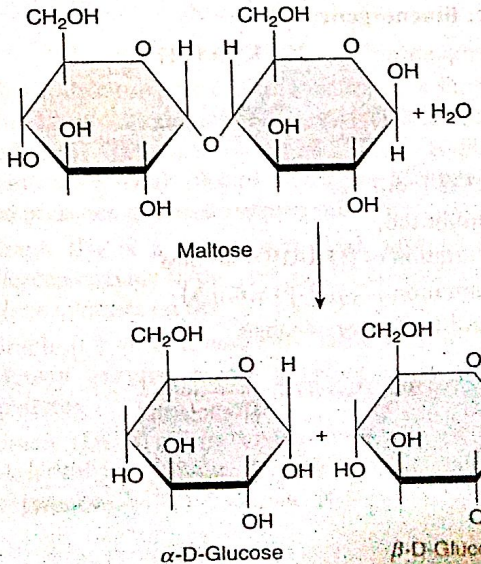
36. Topic: Enzymes–Kinetics

More is the value of r/c, more is the efficiency of binding the ligand by a receptor. On analyzing the plot, it is clear that value of r/c is highest for receptor R1. Thus, receptor R1 has got the highest affinity for the ligand.

Answer (A)

37. Topic: Biomolecules—Structure and Function

Maltose ($C_{12}H_{22}O_{11}$) is a disaccharide composed of two D-glucose residues joined together by $\alpha(1\rightarrow4)$ glycosidic bond. It is found in sprouting grains. It is found less commonly in nature than sucrose and lactose. On hydrolysis, maltose splits into two molecules of glucose.



38. Topic: Biomolecules—Structure and Function

The amino acid side chains in globular proteins are spatially distributed according to their polarities:

- The non-polar residues Val, Leu, Ile, Met, and Phe occur mostly in the interior of a protein, out of contact with the aqueous solvent. The hydrophobic effects that promote this distribution are largely responsible for the three-dimensional structure of native proteins.
- The charged polar residues Arg, His, Lys, Asp, and Glu are usually located on the surface of a protein in contact with the aqueous solvent.
- The uncharged polar groups Ser, Thr, Asn, Gln, and Tyr are usually on the protein surface but also occur in the interior of the molecule. When buried in the protein, these residues are almost always hydrogen bonded to other groups; in a sense, the formation of a hydrogen bond neutralizes their polarity.

Answer (C)

39. Topic: Metabolism—Carbohydrate

During exercise, some of the pyruvate generated by increased glycolytic flux is directed toward oxaloacetate synthesis as catalyzed by pyruvate carboxylase. Pyruvate can accept an amino group from glutamate (a transamination reaction) to generate alanine (the amino acid counterpart of pyruvate) and the citric acid cycle intermediate α-ketoglutarate (the ketone counterpart of glutamate). Both of these mechanisms help the citric acid cycle efficiently catabolize the acetyl groups derived—also from pyruvate—by the reactions of the pyruvate dehydrogenase complex. The end result is increased production of ATP to power muscle contraction.

Answer (C)

40. Topic: Bioenergetics

At, Temperature, $T = 298$ K and pH = 7

$$\underset{44\,mM}{DHAP} \rightleftharpoons \underset{\text{Isomerase}}{\overset{\text{Triose phosphate}}{}} \underset{0\,mM}{G\text{-}3\text{-}P}$$

$R = 8.315\ J\ mol^{-1}\ K^{-1}$

At equilibrium,

Concentration of [DHAP] = 40 mM

Concentration of [G-3-P] = 4 mM

Standard free energy change

$$(\Delta G^{\circ}) = -RT \ln \frac{[\text{Products}]}{[\text{Reactants}]}$$

$$= \frac{-8.315\,J}{K \cdot mol} \times 298K \times \ln \frac{[4mM]}{[40mM]}$$

$$= \frac{8.315\,J}{K \cdot mol} \times 298K \times \ln \frac{[4mM]}{[40mM]}$$

$$= 2477.5 \times (-2.3)$$

$$= 5698.4\ J\ mol^{-1} \approx 5.7\ kJ\ mol^{-1}$$

Answer (5.7 kJ mol^{-1})

41. Topic: Photosynthesis

- Phototropins are blue-light receptors controlling a range of responses that serve to optimize the photosynthetic efficiency of plants. These include phototropism, light-induced stomatal opening, and chloroplast movements in response to changes in light intensity. It contains two light, oxygen, and voltage domains in the N-terminus, which bind a chromophore flavin mononucleotide (FMN) and a serine/threonine kinase domain in the C-terminus, which is required for autophosphorylation of the protein. The action spectrum of plant phototropin is in the UV-A and blue light range (360–500 nm).
- The blue light receptors cryptochromes mediate various light responses in plants. The photoexcited cryptochrome molecules undergo a number of biophysical and biochemical changes, including electron transfer, phosphorylation, and ubiquitination, resulting in conformational changes to propagate light signals. The structure of cryptochrome involves a fold very similar to that of photolyase, with a single molecule of FAD non-covalently bound to the protein. Its activity is fairly uniform throughout its range of maximal response of UV-A and blue light (390–480 nm).
- The phytochrome molecule is the photoreceptor for red light responses. It exists in two forms, Pr and Pfr. The Pr form absorbs at a peak of 666 nm. The Pfr form absorbs at a peak of 730 nm. Phytochrome is a soluble chromoprotein with a molecular mass of 250 kDa it occurs as a dimer made up of two subunits, each of 125 kDa. Each subunit consists of two components — light-absorbing pigment molecule, chromophore and a polypeptide chain, Apo protein. The chromophore is a linear tetrapyrrole similar to phycocyanin termed phytochromobilin and it is a ring attached to the protein through thioether-linkage to a cysteine residue.

Answer (A)

42. Topic: Biological Membrane—Structure

In nature and in the laboratory, a decrease in temperature may cause liquids to solidify and an increase in temperature may cause solids to liquefy. In all instances the lipids are fluid at temperatures at which the organism

normally exists. In general, the higher this temperature, the higher the lipid melting point; the lower the temperature, the lower the lipid melting point. It is well known that straight chain saturated fatty acids have higher melting point that same length unsaturated fatty acids. At low temperatures, the degree of unsaturation of fatty acids is increased through complex biosynthesis pathways. Cold acclimation increases the ratio of unsaturated to saturated fatty acids.

Answer (A)

43. Topic: Biological Membrane—Action Potential

A protoplast when placed in distilled water which acts as hypotonic solution for it, will absorb water by endosmosis, so that the increased volume of water in the cell will increase pressure, making the protoplasm push against the cell wall, a condition known as turgor. They swell quickly and, ultimately cell membrane burst. Thus, the number of intact cells will decrease with time. Thus, graph in option (B) and (C) are wrong.

The protoplasts do not start to bust immediately. For first few minutes, it just absorbs water and swell and when the intake of water exceeds its limit then the cells burst. Thus, graph (A) is correct.

Answer (A)

44. Topic: Biomolecules—Protein Structure and Function

Alpha-Beta domains are the most commonly observed domain structure. In general, they consist of parallel or mixed beta sheets connected and surrounded by alpha helices. Alpha-beta barrels must have at least 8 parallel beta strands (can be more) that form a closed circle and enclose a hydrophobic core. Cross connections between the parallel beta strands are alpha helices as in the beta-alpha-beta motif. All alpha helices are on one side of the beta sheet. The active site of all alpha-beta barrels is found in a pocket formed when carboxy ends of parallel strands loop to connect to amino ends of adjacent alpha helices.

Answer (C)

45. Topic: Biomolecules—Protein Structure and Function

Levinthal's paradox is an apparent contradiction between the number of possible conformations for a protein chain and the fact that proteins can fold to their native conformation quickly. The paradox is resolved by thermodynamic and kinetic principles which create a folding pathway for the unfolded protein. The paradox is reopened by intrinsically disordered regions of proteins which are inherently random. Multiple techniques like NMR, circular dichroism, X-ray crystallography and cryo-EM are used together to confirm the protein structure.

Answer (C)

46. Topic: Enzymes—Kinetics

Katal (kat) is defined as the amount which will catalyze the transformation of one mole of substance per second.

Given,

Concentration of substrate = 450 μ moles

$= 450 \times 10^{-6}$ moles

Time = 1 min = 60 s

We know,

$$\text{Enzymatic activity, } E_a = \frac{\text{moles}}{\text{time}}$$

$$= \frac{420 \times 10^{-6}}{60} = 7 \times 10^{-6} \text{ Katal}$$

Answer (7×10^{-6} Katal)

47. Topic: Biomolecules—Protein Structure and Function

The charged polar residues Arg, His, Lys, Asp and Gl are usually located on the surface of a protein in contact with the aqueous solvent. In contrast, amino acids with non-polar side groups avoid water and aggregate to form the water insoluble core of proteins.

Answer (C)

48. Topic: Enzymes—Kinetics

In uncompetitive inhibition, the inhibitor binds directly to the enzyme–substrate complex but not to the free enzyme. An uncompetitive enzyme inhibitor affects catalytic activity such that both that apparent K_M and the apparent V_{max} decrease. The Michaelis–Menten equation for uncompetitive inhibition and the equation for its double-reciprocal plot are:

$$V_o = \frac{V_{max}[S]}{K_M + \alpha'[S]} = \frac{(V_{max}/\alpha')[S]}{(K_M/\alpha') + [S]}$$

$$\frac{1}{v_o} = \frac{K_M}{V_{max}}\frac{1}{[S]} + \frac{\alpha'}{V_{max}}$$

The double-reciprocal plot consists of a family of parallel lines with slope K_M/V_{max}, $1/V_o$ intercepts of α'/V_{max}, and $1/[S]$ intercepts of $-\alpha'/K_M$. Thus graph (A) is a Lineweaver–Burk plot of Michaelis–Menten enzyme in the presence of uncompetitive inhibitior.

Graph (B) is a Lineweaver–Burk plot of Michaelis–Menten enzyme in the presence of competitive inhibitor. Lines intersect on the $1/V_o$ axis at $1/V_{max}$.

Graph (C) is a Lineweaver–Burk plot of Michaelis–Menten enzyme in the presence of non-competitive inhibitor.

Graph (D) is a Lineweaver–Burk plot of Michaelis–Menten enzyme in the presence of mixed inhibitor. Lines intersect to the left of the $1/V_o$ axis.

Answer (A)

49. Topic: Enzymes—Kinetics

A competitive inhibitor therefore reduces the concentration of free enzyme available for substrate binding.

Given, $K_i = 2.5\ \mu M$

Concentration of inhibitor, $[I] = 5\ \mu M$

K_M^{app} in presence of inhibitor $= 30\ \mu M$

We know,

$$K_M^{app} = K_M\left[1+\frac{I}{K_i}\right]$$

K_M is measured in the absence of inhibitor.

$$K_M = \frac{K_M^{app}}{\left[1+\frac{I}{K_i}\right]}$$

Putting the values in the formula we get:

$$K_M = \frac{30}{\left[1+\frac{5}{2.5}\right]} = \frac{30}{3} = 10\,\mu M$$

Answer (10 μM)

50. Topic: Bioenergetics

Given,

Standard free energy change (ΔG°) for ATP hydrolysis $= -30\ kJ\ mol^{-1}$

In vivo concentrations of [ATP] = 7.09 mM

[ADP] = 1.04 mM

$[P_i] = 7.9$ mM

Temperature $= 37^\circ C = 37 + 273 = 310\ K$

Reaction: $ATP \rightleftharpoons ADP + P_i$

$$K = \frac{[ADP][P_i]}{[ATP]} = \frac{[1.04][7.9]}{[7.09]}$$

$$= 1.04\ mM = 1.04 \times 10^{-3}\ M$$

Now, we know,

$$\Delta G = \Delta G^\circ + RT \ln K$$

$$= -30 + 0.008314 \times 310 \ln[1.04 \times 10^{-3}]$$

$$= -30 + 2.57 \times -6.9$$

$$= -30 - 17.7$$

$$= -47.7\ kJ\ mol^{-1}$$

Answer ($-47.7\ kJ\ mol^{-1}$)

51. Topic: Electron Transport Chain

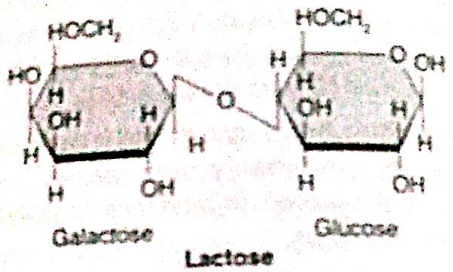
ATP synthesis occurring due to oxidation-reduction in the proton gradient formed across the mitochondrial membrane. Higher concentration of protons [H^+] is present in the thylakoid lumen (interior space of thylakoid). The free energy of the proton gradient is tapped by chloroplast ATP synthase.

The light reactions of photosynthesis establish a proton gradient across the thylakoid membrane that leads to ATP formation. Removal of H^+ from the stroma and addition of H^+ to the thylakoid lumen leads to the formation of proton gradient. Synthesis of ATP in mitochondria takes place in its inner membrane where an ionic gradient is established. Oxidative phosphorylation is the process in which ATP formation is driven by energy released from electron removed during substrate oxidation.

Answer (B)

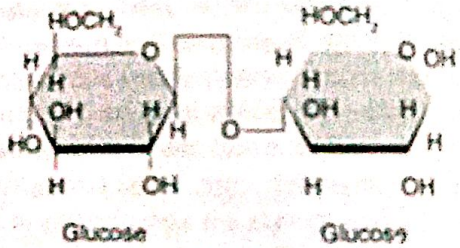
52. Topic: Biomolecules—Carbohydrate Structure and Function

A disaccharide is a molecule formed from the condensation of two monosaccharides by dehydration synthesis. Their molecules yield two molecules of the same or different monosaccharides when hydrolyzed. The systematic name for lactose is o-β-D-galactopyranosyl-(1→4)-D-glucopyranose. The glycosidic bond links C1 of the β-anomer of galactose to O4 of glucose.



Lactose

Cellobiose is a disaccharide consists of two β-glucose residues that are linked by $\beta(1\rightarrow4)$ glycosidic bonds.

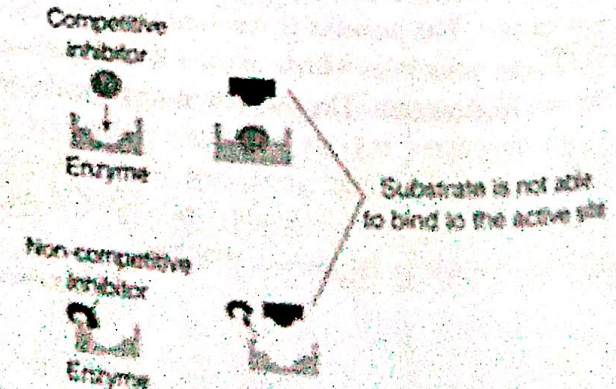


Cellulose

Answer (D)

53. Topic: Enzymes—Kinetics

A transition state is observed when the reactants are at the crest of the hump. At this state, they are ready to be converted to products. As the transition state is converted to products, the affinity of the enzyme for the bound molecule(s) decreases as the products are unable to bind to the buttressing group and get expelled.



(a) Competitive inhibition

(b) Non-competitive inhibition

- In competitive inhibition, the inhibitor binds directly to enzyme substrate complex but not to the free enzyme.
- Hence, even if a large amount of substrate added to an enzyme, it cannot overcome uncompetitive inhibition.
- In non-competitive inhibition (also called mixed inhibition), both enzyme and enzyme-substrate complex bind inhibitor. The inactivator (irreversible inhibitor) reduces the concentration of functional enzyme at all substrates concentration. Hence, V_{max} decreases and K_M remains unchanged.

Answer (C)

54. Topic: Biological Membrane—Action Potential

$$\Delta G^\circ = RT \ln \frac{[Na^+]_{in}}{[Na^+]_{out}} + Z \cdot F \cdot \psi$$

At equilibrium, $\Delta G = 0$

$$RT \ln \frac{[Na^+]_{in}}{[Na^+]_{out}} = -Z \cdot F \cdot \psi$$

Given, Equilibrium potential, $\psi = 55mV = 55 \times 10^{-3}$ V

Gas constant, $R = 1.98$ cal.mol^{-1}.K^{-1}

Temperature, $T = 37 + 273 = 310$ K

Faraday constant $F = 23062$ cal V^{-1} mol^{-1}

Inside concentration of Na^+, $[Na^+]_{in} = 20mM$

Z for $Na^+ = 1$

Putting all these in equation above we get:

$$\ln \frac{20}{[Na^+]_{out}} = -\frac{1 \times 23062 \times 55 \times 10^{-3}}{1.98 \times 310}$$

$$\ln \frac{20}{[Na^+]_{out}} = -2.07 \text{ or } \ln \frac{[Na^+]_o}{20} = 2.07$$

$$\frac{[Na^+]_o}{20} = e^{2.07}$$

$$\frac{[Na^+]_o}{20} = 7.942$$

$$[Na^+]_o = 158.49\,mM$$

Answer (158.49 mM)

55. Topic: Enzymes—Kinetics

Molecular mass of type II dehydroquinase = 18kDa
$= 18 \times 10^3$ Da
$= 18 \times 10^3$ g

Amount of enzyme added = 1.8 µg

To find concentration of enzyme $[E_0]$,

18×10^3 g of enzyme = 1 mol

$$1.8\ \mu g \text{ of enzyme} = \frac{1.8 \times 10^{-6}}{18 \times 10^3} = 10^{-10} \text{ mol}$$

$$\therefore [E_0] = 10^{-10} \text{ mol}$$

V_{max} of the enzyme = 0.0134 µmol min^{-1}

$$= \frac{0.0134 \times 10^{-6}}{60} \text{ mol s}^{-1}$$

We know,

$$V_{max} = k_{cat}[E_0]$$

$$k_{cat} = \frac{V_{max}}{[E_0]} = \frac{0.0134 \times 10^{-6}}{60 \times 10^{-10}} = 2.23 s^{-1}$$

Now, to calculate turnover number,

K_M for the substrate = 25 µM = 25×10^{-6} M

$$\frac{k_{cat}}{K_M} = \frac{2.23}{25 \times 10^{-6}} = 8.93 \times 10^4 M^{-1} s^{-1}$$

Answer ($8.93 \times 10^4 M^{-1}s^{-1}$)

56. Topic: Biomolecules—Structure and Function

For any peptide amino side lies at left side from where sequence started and carboxyl group at right side where sequence ends.

- Option A: Let us assume the amino acid sequence to be RSYRVA. On its trypsin digestion that cleaves at carboxyl side of R as shown:

$$R^{\downarrow}SYR^{\downarrow}VA$$

Three fragments (R), (S, Y, R) and (V, A) will be generated which are clearly not satisfying the condition given in the question. Thus, this option is wrong.

- Option B: Let us assume the amino acid sequence to be AVRYSR.

(a) On its digestion with trypsin that cleaves at carboxyl side of R as shown:

$$AVR^{\downarrow}YSR^{\downarrow}$$

Two fragments (A,V, R) and (Y, S, R) will be generated which is satisfying the condition given in question.

(b) On its digestion with chymotrypsin that cleaves at carboxyl side of Y as shown:

$$AVRY^{\downarrow}SR$$

Two fragments (A,V, R,Y) and (S, R) will be generated which is satisfying the condition given in question.

(c) On its digestion with carboxypeptidase that cleaves at amino side of C (except R and K), no cleavage product will be formed, as its C-terminal has R which is an exception.

AVRYSR

Thus, this the correct option.

- Option C: Let us assume the amino acid sequence to be SRYVAR. On its trypsin digestion that cleaves at carboxyl side of R as shown:

$$SR^{\downarrow}YVAR^{\downarrow}$$

Two fragments (S, R) and (Y, V, A, R) will be generated which are clearly not satisfying the condition given in the question. Thus, this option is wrong.

- Option D: Let us assume the amino acid sequence to be SVRRYA. On its trypsin digestion that cleaves at carboxyl side of R as shown:

$$SVR^{\downarrow}R^{\downarrow}YA$$

Three fragments (S, V, R), (R) and (Y, A) will be generated which are clearly not satisfying the condition given in the question. Thus, this option is wrong.

Answer (B)

57. Topic: Enzymes—Mechanism of Action

Activation energy is the minimal kinetic energy needed for a reactant to undergo a chemical reaction. It is the difference in energy between the substrate and the transition state. Apart from bringing the substrates in proper orientation, enzymes also lower the activation energy of a chemical reaction by decreasing the "randomness" of the collisions between molecules.

The enzyme action can thus be summarized as:

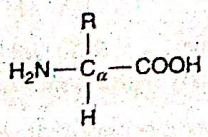
$$E + S \rightarrow ES \rightarrow EP \rightarrow E + P$$

An enzyme is responsible for providing a lower energy pathway from substrate to product. It, however, does not affect the overall free energy change for the reaction.

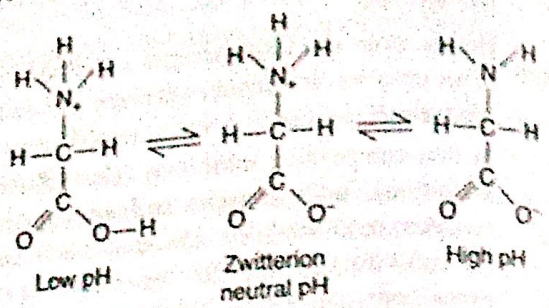
Answer (B)

58. Biomolecules—Protein Structure and Function

All proteins are composed of 20 standard amino acids. The common amino acids are known as α-amino acids as they have a primary amino group ($—NH_2$) as a substituent of the α-carbon atom, the carbon next to the carboxylic acid group ($—COOH$). The 20 amino acids differ in the structures of their side chains (R groups). The amino and carboxylic groups of amino acids can readily ionize such molecules that carry charged groups of opposite polarity are called dipolar ions or zwitterions.



The α-amino acids generally have two, of for those ionizable side chains, three acid-base groups. When the pH is very low, these groups are fully protonated. When the pH is very high, these are unprotonated.



Answer (A)

59. Topic: Biomolecules—Structure and Function

Two cysteine residues can be linked by a disulfide bond to form cystine. Thus, four cysteine residues can form two disulfide bonds. Let us assume that in N number of different ways four cysteine residues can form two disulphide bonds:

$$N = \frac{n!}{(n-2p)!\,p!\,2^p}$$

where, n = number of cysteine residue = 4

p = number of disulphide bond to be formed = 2

$$N = \frac{4!}{(4-2\times2)!\,2!\,2^2}$$

$$= \frac{4\times3\times2}{(0)!\times2\times4} = 3$$

Answer (3)

60. Topic: Enzymes—Kinetics

In 1913, Leonor Michaelis and Maud Menten derived a mathematical relationship between substrate concentration and the velocity of enzyme reactions. They measured the amount of product formed (or substrate consumed) in a given time period and gave the following equation, also known as Michaelis–Menten equation.

$$V = V_{max}\frac{[S]}{[S]+K_M}$$

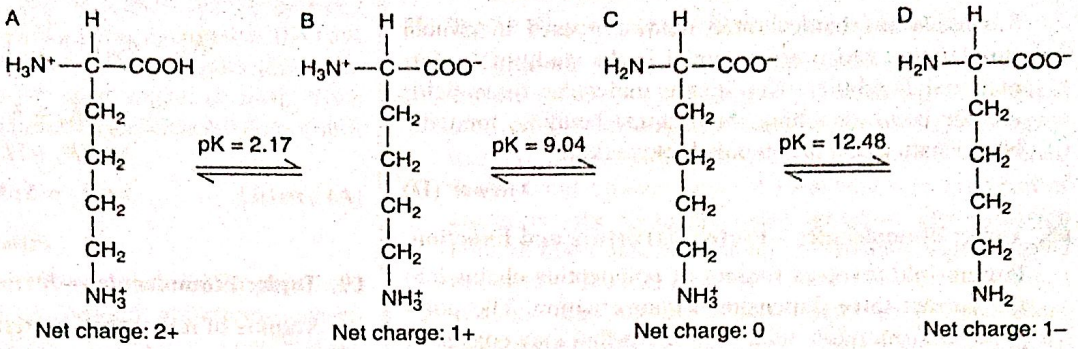
where, V = initial velocity of the reaction; V_{max} = maximum rate of reaction; K_M = Michaelis-Menten constant and $[S]$ = substrate concentration.

According to the equation, when the substrate concentration $[S]$ is set at a value equivalent to K_M, then the velocity of the reaction (V) becomes equal to $V_{max}/2$, or one-half the maximal velocity. Thus, $K_M = [S]$, when $V = V_{max}/2$. Michaelis constant (K_M) is the substrate concentration when the reaction velocity is one-half of V_{max}. It is constant for a given enzyme and is independent of substrate or enzyme concentration.

Answer (B)

61. Topic: Biomolecules—Structure and Function

The five complex amino acids include Glutamic acid (Glu), Aspartic acid (Asp), Lysine (Lys), Arginine (Arg) and Histidine (His). Each of these has 3 ionizable groups. The three ionization points for Lys is shown below:



Answer (D)

62. Topic: Biomolecules—Protein Structure and Function

Chemical composition of protein = $CH_{1.5}O_{0.3}N_{0.3}S_{0.004}$

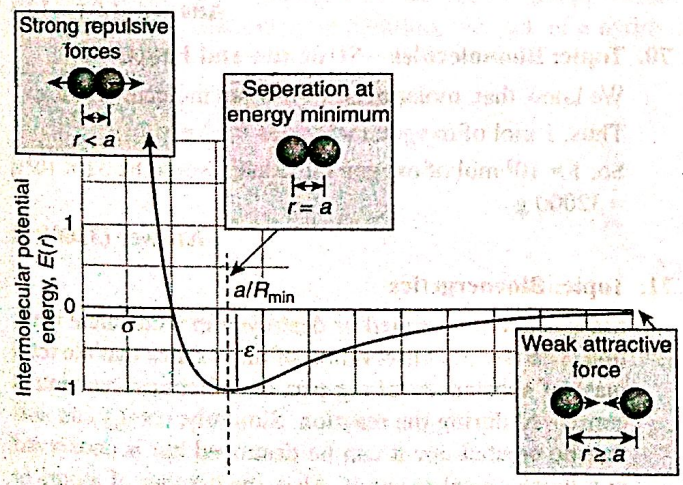
$$\text{Mass of protein} = (1\times12)+(1.5\times1)+(0.3\times16)+(0.3\times14)+(0.004\times32)$$
$$= 12+1.5+4.8+4.2+0.128$$
$$= 22.628\,\text{g}$$

$$\text{Mass \% of carbon} = \frac{1\times12}{22.628}\times100$$
$$= 53.03\%$$

Answer (53.03%)

63. Topic: Biomolecules—Structure and Function

Lennard-Jones potential is the most known realistic model of a two-particle interaction.



$$V_{LJ}(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

$$V_{LJ}(r) = \varepsilon\left[\left(\frac{R_{min}}{r}\right)^{12} - 2\left(\frac{R_{min}}{r}\right)^{6}\right]$$

Here, constant, ε = the interaction strength (the depth of the attractive well)

σ = the range of interaction (the finite distance, where the interaction energy is zero).

R_{min} = the distance where the potential reaches a minimum, i.e., the equilibrium position of the two particles, here it is equal to a.

Thus,

$$V_{LJ}(r) = \varepsilon\left[\left(\frac{a}{r}\right)^{12} - 2\left(\frac{a}{r}\right)^{6}\right]$$

The first part of the equation, $(a/r)^{12}$ describes the repulsive forces between particles, while the latter part of the equation, $(a/r)^{6}$ denotes attraction. Like the bonding potential energy, the stability of an arrangement of atoms is a function of the Lennard-Jones separation distance.

- As the separation distance decreases below equilibrium ($r < a$), the potential energy gradually becomes positive (indicating a repulsive force). Such a large potential energy is energetically unfavorable, as it indicates an overlapping of atomic orbitals.
- At long separation distances ($r \geq a$), the potential energy is negative and approaches zero as the separation distance increases to infinity (indicating an attractive force). This indicates that at long-range distances, the pair of atoms or molecules experiences a small stabilizing force.
- As the separation between the two particles reaches a distance slightly greater than σ ($r = a$), the potential energy reaches a minimum value (indicating zero force). At this point, the pair of particles is most sta-

ble and will remain in that orientation until an external force is exerted upon it.

Answer (D)

64. Topic: Biomolecules—Structure and Function

Molecular mechanics force fields are used in protein simulations, which are essential in the study of conformational flexibility. The simple molecular force fields include bond stretching, bond angle bending, torsional bond rotation and non-bonded interactions.

Answer (D)

65. Topic: Biomolecules—Protein Structure and Function

Protein fold involves folding of polypeptide chains into the correct three-dimensional conformation. The polypeptides from multi subunit proteins that may consist of identical or non-identical polypeptide chains.

Answer (B)

66. Topic: Respiration

Glycolysis occurs in the cytoplasm and is not associated with any organelles or membrane structures and it also does not require oxygen.

Answer (D)

67. Topic: Bioenergetics

From the equation:

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$$

It is clear that one mole of glucose on complete conversion produces 2 moles of ethanol.

Molecular weight of glucose = $6 \times 12 + 12 \times 1 + 6 \times 16$
= 180 g

Molecular weight of ethanol = $2 \times 12 + 5 \times 1 + 1 \times 16 + 1 \times 1 = 46$ g

Thus, 180 g of glucose produces = 2×46
= 92 g of ethanol

200 g of glucose will produce = $92 \times 200 / 180$
= 102.22 g ethanol

Answer (102.22 g)

68. Topic: Enzymes—Kinetics

Given, Concertation of Inhibitor $[I] = 10$ nM

$$\text{Slope of Lineweaver-Burk plot } (S) = \frac{K_M}{V_{max}}$$

$$\text{New slope } (S') = \frac{K'_M}{V_{max}}$$

As the velocity remains constant.

As per the question,

$$S' = 3S$$

$$\Rightarrow \frac{K'_M}{V_{max}} = 3\frac{K_M}{V_{max}}$$

$$\Rightarrow K'_M = 3K_M$$

Now, $K'_M = K_M\left[1+\frac{[I]}{K_I}\right]$

$$3K_M = K_M\left[1+\frac{[I]}{K_I}\right] \quad \text{[From equation (i)]}$$

$$\Rightarrow 3K_I = K_I + [I]$$

$$\Rightarrow 2K_I = [I] = 10$$

$$\Rightarrow K_I = 5\text{ nM}$$

Answer (5 nM)

69. Topic: Biomolecules—Structure and Function

Volume of rod-shaped bacterial cell = $\pi r^2 h$

$= 3.14 \times 0.5 \times 10^{-6} \times 0.5 \times 10^{-6} \times 2 \times 10^{-6}$

$= 1.57 \times 10^{-18}$ m^3

Given, Density of cell = Density of water

$= 1000$ g m^{-3}

Mass of cell = Density × Volume

$= 1000$ g m$^{-3} \times 1.57 \times 10^{-18}$ m^3

$= 1.57 \times 10^{-12}$ g

Mass of Proteins = 15% of cell mass

$= 15/100 \times 1.57 \times 10^{-12}$

$= 23.55 \times 10^{-14}$ g

Given, 1.6×10^{-24} g = 1 Da

23.55×10^{-14} g $= 23.55 \times 10^{-14} / 1.6 \times 10^{-24}$

$= 14.7185 \times 10^{10}$ Da

50 kDa is mass of 1 protein.

14.7185×10^{10} Da would be mass of

$14.7185 \times 10^{10} / 50 \times 10^3 = 2.94 \times 10^6$ Proteins

Answer (2.94×10^6)

70. Topic: Biomolecules—Structure and Functions

We know that, molar mass of oxygen molecule is 32 g

Thus, 1 mol of oxygen molecules = 32 g

So, 1×10^3 mol of oxygen molecules would be 32×10^3 g
= 32000 g

Answer (32000 g)

71. Topic: Bioenergetics

Mass cannot be created or destroyed in a chemical reaction. The law of conservation of mass states that the total mass of substances taking part in a chemical reaction is conserved during the reaction. Similarly, energy can neither be created nor it can be destroyed but is conserved in a biochemical reaction. Also, the number of atoms in a reaction remains the same. Moles are conserved only when there is no reaction. Volume is not conserved.

Answer (B)

72. Topic: Enzymes—Kinetics

Enzymes affect the free energy graph by lowering the activation barrier, or the amount of energy needed for the reaction to occur. They do this by stabilizing the transition state, or the state of highest energy between reactants and products. An enzyme cannot alter the laws of thermodynamics and consequently cannot affect the equilibrium constant in any way, shape, or form. They just accelerate the attainment of equilibria but do not shift their positions.

Answer (A)

73. Topic: Enzymes—Kinetics

Specific activity is a ratio of the enzymatic reactions of a particular protein to the total amount of proteins, quantifying a protein can be followed throughout a purification. The equation for specific activity can be modeled as:

$$\text{Specific activity} = \frac{\text{Total enzyme activity}}{\text{Total protein}}$$

Protein concentration = 10 mg mL^{-1}

10 µL = 10×10^{-3} mL of this sample has = $10 \times 10^{-3} \times$ 10 mg = 0.1 mg

Enzyme activity = 5 µmol min^{-1}

$$\therefore \text{Specificactivity} = \frac{5\ \mu\text{mol min}^{-1}}{0.1\ \text{mg}} = 50\ \mu\text{mol min}^{-1}\ \text{mg}^{-1}$$

Answer (50 µmol min^{-1} mg^{-1})

74. Topic: Biomolecules—Protein Structure and Function

The structural integrity of an α-helix is in part dependent on correct steric configuration. Amino acids whose R-groups are too large (tryptophan, tyrosine) or too small (glycine) destabilize α-helices. Proline also destabilizes α-helices because of its irregular geometry; its R-group bonds back to the nitrogen of the amide group, which causes steric hindrance. In addition, the lack of a hydrogen on proline's nitrogen prevents it from participating in hydrogen bonding. Thus, proline in alpha helix cause a kink in the helix.

Answer (A)

75. Topic: Biomolecules—Structure and Function

A hydrogen bond is a partially electrostatic force of attraction between a hydrogen (H) atom which is bound to a more electronegative atom or group, such as nitrogen (N), oxygen (O), or fluorine (F)—the hydrogen bond donor—and another adjacent atom bearing a lone pair of electrons—the hydrogen bond acceptor. The hydrogen bond is often described as an electrostatic dipole-dipole interaction. It is directional. It is a weak bond, but it is stronger than van-der Waals forces. In the secondary structure of proteins, hydrogen bonds form between the backbone oxygens and amide hydrogens.

Answer (C)

76. Topic: Transport Processes

Energy required for active transport,

$$\Delta G = RT \ln \frac{[\text{Solute}]_{in}}{[\text{Solute}]_{out}}$$

Given, $R = 1.987$ cal mol^{-1} K^{-1}

$T = 37°C = 37 + 273 = 310$ K

Difference in concentrations of an uncharged solute between two compartments is 1.6-fold

Thus, $\frac{[\text{Solute}]_{in}}{[\text{Solute}]_{out}} = 1.6$

$$\Delta G = 1.987 \times 310 \times \ln 1.6$$
$$= 615.97 \times 0.47$$
$$= 289.51\ \text{cal mol}^{-1}$$

Answer (289.51 cal mol^{-1})